

# Technical Document #849: Data Engineering - Comprehensive Overview

## Technical Document #849: Data Engineering - Comprehensive Overview

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

### Key Takeaways:

- Stream processing handles continuous data streams in real-time.
- Partitioning divides data into smaller, more manageable pieces.
- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.